

Qatar-1b

R-2489

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_260705 · created 2026-08-24T01:07:09+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2461226.9111 +/- 0.0034 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.163 +/- 0.057
Transit depth (Rp/Rs)^2	2.66 +/- 1.84 [%]
Orbital Inclination (inc)	79.65 +/- 2.87
Airmass coefficient 1 (a1)	2.32 +/- 1.22
Airmass coefficient 2 (a2)	-0.64 +/- 1.6
Scatter in the residuals of the lightcurve fit is	56.18 %
Best Comparison Star	None
Optimal Aperture	0.87
Optimal Annulus	6.75
Transit Duration (day)	0.027 +/- 0.028

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.163 $\pm$ 0.057 vs 0.1463 (deviation 0.28 σ)
Duration fit vs published	0.027 d vs 0.0692 d (deviation 1.45 σ)
Mid-time O-C	2.9 min
Depth SNR	2.7
Residual scatter	56.18 %

Light curve

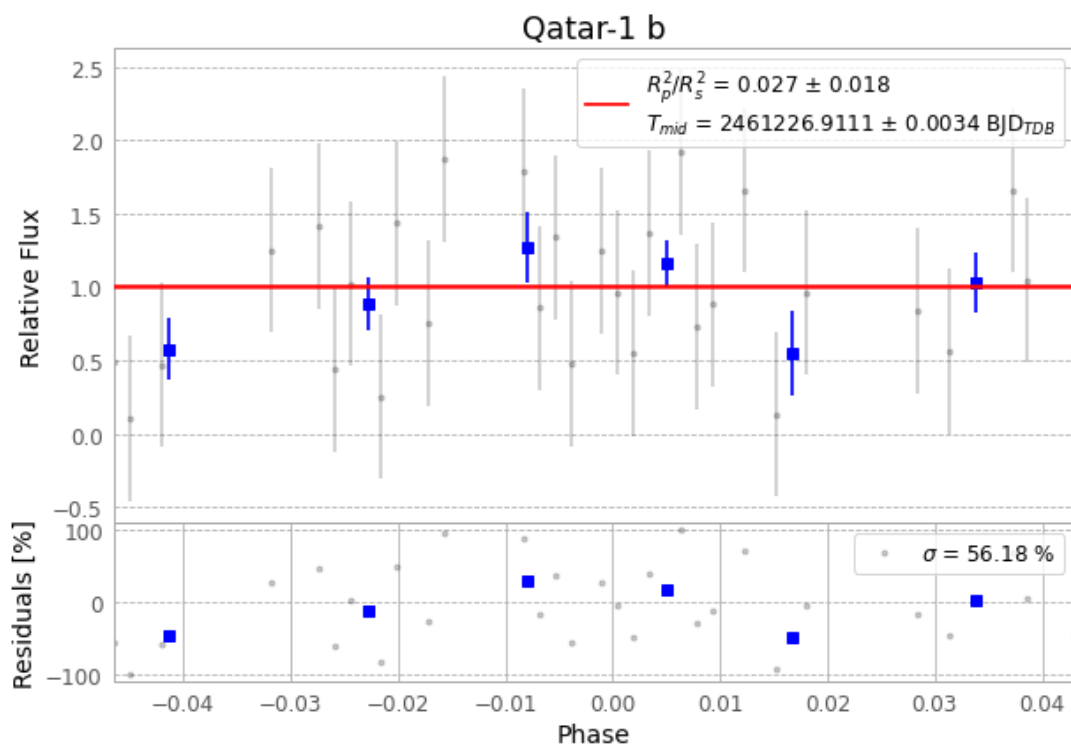


Figure 1 — FinalLightCurve_Qatar-1 b_2026-07-05.png (SHA-256 da326e57af551fb6...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [318, 134], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 15bf79e63cfc10b4...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

138 FITS frames (59 MB), Qatar-1260705075757.FITS.gz → Qatar-1260705120409.FITS.gz. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-260705.log (710890027f04...), FinalLightCurve_Qatar-1 b_2026-07-05.png (da326e57af55...), FinalLightCurve_Qatar-1 b_2026-07-05.csv (a48d9bbdd2d9...)

Compute resources

Wall time 148.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-24 01:07Z.